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STUDY MATERIALS



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APJ ABDULKALAM TECHNOLOGICAL UNIVERSITY
FIRST SEMESTER B-TECH DEGREE EXAMINATION DECEMBER 2019
COURSE CODE: EST 130
COURSE NAME: BASICS OF ELECTRICAL AND ELECTRONICS
ENGINEERING
PART II: BASIC ELECTRONICS ENGINEERING
(2019 SCHEME)

PART A

Answer all question, each carry 4 marks

1. What are the different types of capacitors? Give any two applications of capacitors.

A capacitor is a physical device which is capable of storing energy by virtue of a voltage existing across it. The different types of capacitors are fixed and variable type of capacitors.

Fixed types of capacitors are

- Paper capacitor
- Mica capacitors
- Ceramic capacitors
- Electrolytic capacitors

Variable capacitors are

- Air gang capacitors and
- Trimmer capacitors

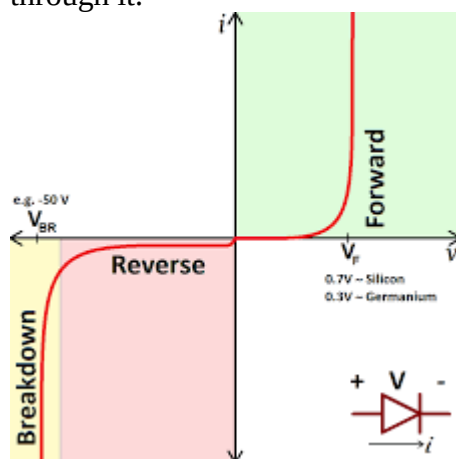
Applications of Capacitors

Energy storage

Power Factor correction

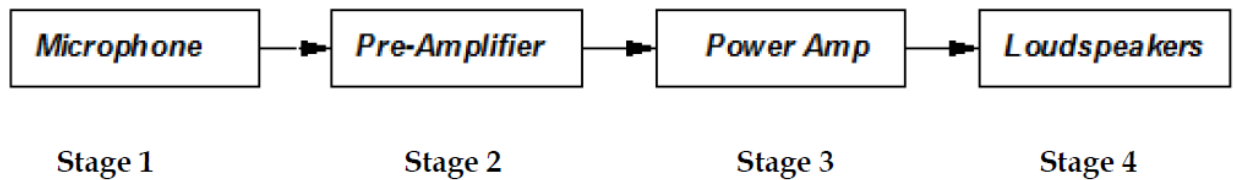
2. Describe the forward Characteristics of a diode.

When positive terminal of the external supply is connected to P region and negative terminal to N region, the PN junction is said to be forward biased. Under forward biased condition the PN region offers a very low resistance and a large amount of current flows through it.



3. Draw the block diagram of Public Address system and write the role of each block.

A simple public address system (or PA system) is shown in the following block diagram.



Stage 1: Microphone (Transducer)

The microphone converts sound waves into electrical signals that can be processed by the rest of the system. It is important that the microphone creates a faithful reproduction of the sound wave as an electrical signal – no distortion!

Stage 2: Pre-Amplifier

Its purpose is to take the small electrical signals from the microphone and increase the amplitude of the signal voltage.

Stage 3: Power Amplifier

The power amplifier takes this enlarged voltage signal, and boosts the current so that it is strong enough to drive the loudspeaker.

Stage 4: Loud Speaker

The loudspeaker is the final part of the system where the electrical signal is transformed back into a sound wave.

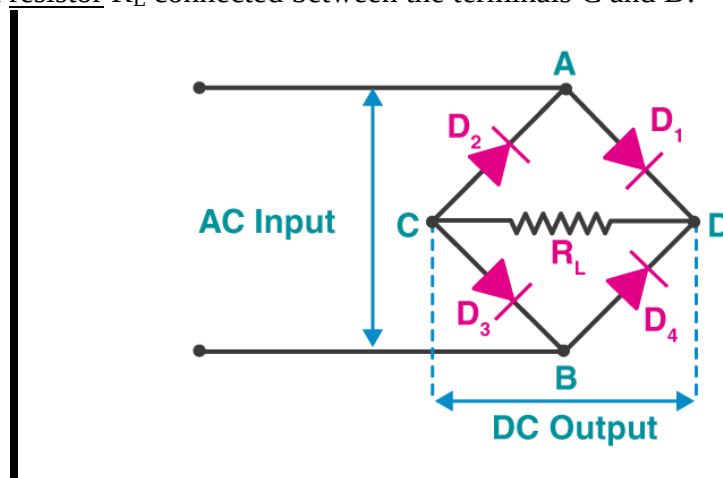
If the system has carried out its function correctly, the emerging sound wave will be an undistorted but amplified version of the original.

4. Explain the working of a bridge rectifier.

Rectifier: A rectifier is a circuit that converts AC into pulsing DC. It uses unidirectional conducting devices like PN diodes. Rectifiers are classified into two types based on the conduction of AC input. They are: (a) Half wave rectifier (HWR). (b) Full wave rectifier (FWR).

Bridge Rectifiers are circuits that convert alternating current (AC) into direct current (DC) using diodes arranged in the bridge circuit configuration. Bridge rectifiers typically comprise of four or more diodes. The output wave generated is of the same polarity irrespective of the polarity at the input.

Bridge rectifiers are in the same class of electronics as half-wave rectifiers and full-wave rectifiers. Figure 1 shows such a bridge rectifier composed of four diodes D_1 , D_2 , D_3 , and D_4 in which the input is supplied across two terminals A and B in the figure while the output is collected across the load resistor R_L connected between the terminals C and D.



Now consider the case wherein the positive pulse appears at the AC input i.e. the terminal A is positive while the terminal B is negative. This causes the diodes D_1 and D_3 to get forward biased and at the same time, the diodes D_2 and D_4 will be reverse biased.

5. Explain the concept of cell in cellular communication.

In mobile communication, the coverage area is divided into smaller areas which are each served by its own base station. These smaller areas are called cells. Spatially reusing the available spectrum so that the same spectrum can support multiple users separated by a distance is called frequency reuse. One of the important features of a cellular system is its ability to transfer calls that are already in progress from one cell-site controller to another as the mobile unit moves from cell to cell within the cellular network. The base station transfer includes converting the call to an available channel within the new cell's allocated frequency subset. The transfer of a mobile unit from one base station's control to another base station's control is called a *Handoff* (or *Handover*).

PART B

Answer one full question from each module , each question carry 10 marks

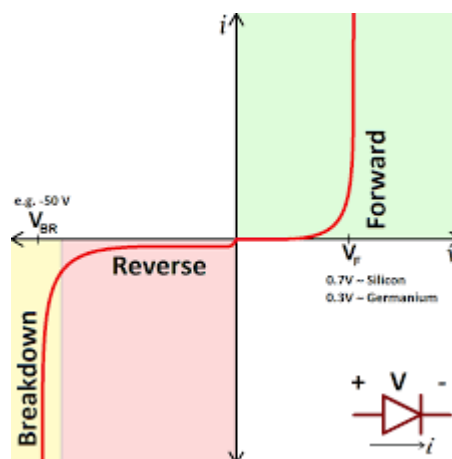
6. a) Explain the formation of potential barrier in a PN junction diode. (4)

The region around the junction from which the mobile charge carriers (electrons and holes) are depleted is called as depletion region. since this region has immobile ions, which are electrically charged , the depletion region is also known as space charge region. Because of the oppositely charged ions present on both sides of PN junction an electric potential is established across the junction even without any external voltage source which is termed as barrier potential.

6 b) What do you understand by Avalanche breakdown? Draw and explain the reverse characteristics of a diode.

When bias is applied , thermally generated carriers which are already present in the diode acquire sufficient energy from the applied potential to produce new carriers by removing valence electron from their bonds. These newly generated additional carriers acquire more energy from the potential and they strike the lattice and create more number of free electrons and holes. This process goes on as long as bias is increased and the number of free carriers get multiplied. This process is termed as avalanche multiplication. Thus the break down which occur in the junction resulting in heavy flow of current is termed as avalanche break down.

When positive terminal of the external supply is connected to P region and negative terminal to N region ,the PN junction is said to be forward biased. under forward biased condition the PN region offers a very low resistance and a large amount of current flows through it.



7. Explain the working of an NPN Transistor. Describe with suitable sketches the input and output characteristics of the input and output characteristics of an NPN transistor.

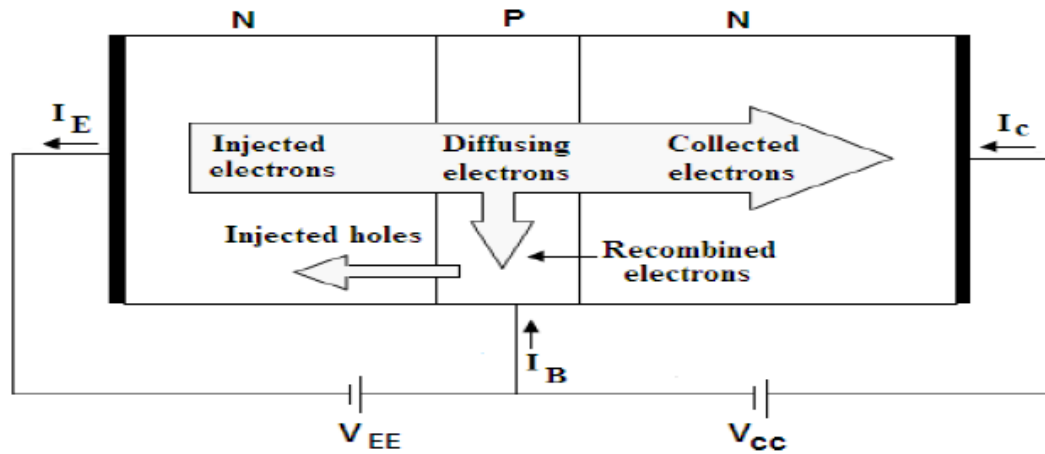


Fig: Operation of NPN Transistor

The current that flows across the emitter-base junction is known as emitter current I_E , which is the sum of electron current and hole current. The direction of I_E is out of the emitter lead, which is in the direction of hole current and opposite to the direction of the electron current. However, the electron component is much larger than the hole component, the emitter current is dominated by the electron component. Thus,

$$I_E = I_{nE} + I_{pE}$$

A few electrons, which are unable to reach the collector-base junction, will recombine with holes, which are majority carriers in the base. It accounts for a small base current (I_B) that is supplied from the base contact.

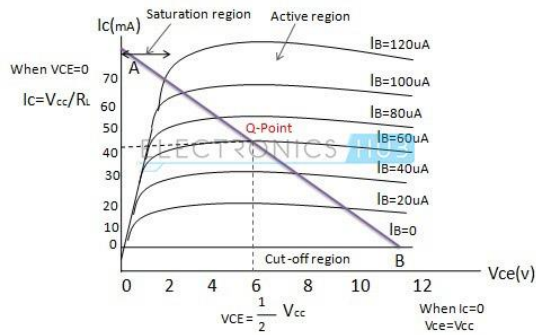
Most of the diffusing electrons will reach the boundary of the collector-base depletion region. Because the collector is more positive than base, the diffused electrons will be swept across the depletion region into the collector. They will be collected to constitute the collector current (I_C). Therefore,

$$I_C = I_{nC}$$

The emitter current is the sum of the collector and the base current. Hence

$$I_E = I_B + I_C$$

The family of output characteristics curves of a bipolar transistor is given below. The curves show the relationship between the collector current (I_C) and the collector-emitter voltage (V_{CE}) with the varying of base current (I_B). We know that the transistor is 'ON' only when at least a small amount of current and small amount of voltage is applied at its base terminal relative to emitter otherwise the transistor is in 'OFF' state.



The collector current (I_C) is mostly affected by the collector voltage (V_{CE}) at 1.0V level but this I_C value is not highly affected above this value. Already we know that the emitter current is the sum of base and collector currents. i.e. $I_E = I_C + I_B$. The current flowing through the resistive load (R_L) is equal to the collector current of the transistor. The equation for the collector current is given by,

$$I_C = (V_{CC} - V_{CE}) / R_L$$

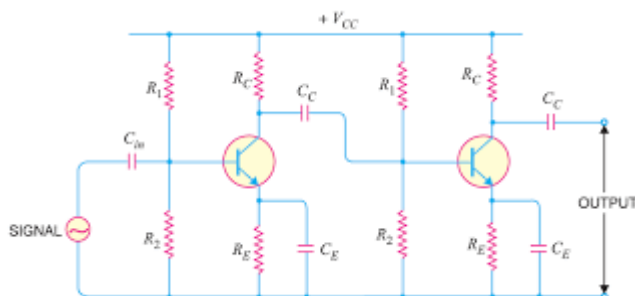
The straight line indicates the 'Dynamic load line' which is connecting the points A (where $V_{CE} = 0$) and B (where $I_C = 0$). The region along this load line represents the 'active region' of the transistor.

The common emitter configuration characteristics curves are used to calculate the collector current when the collector voltage and base current is given. The load line (red line) is used to determine the Q-point in the graph. The slope of the load line is equal to the reciprocal of the load resistance. i.e. $-1/R_L$.

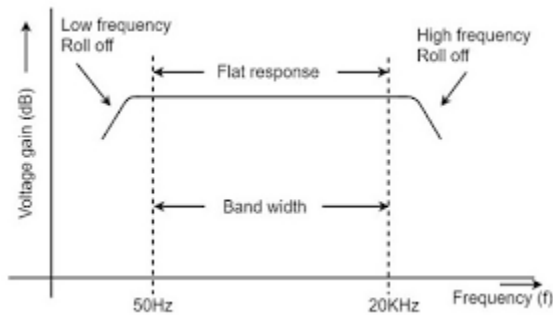
8. Draw the circuit diagram of an RC coupled amplifier and explain its frequency response.

The input capacitor C_{in} couples a.c signal voltage to base of the transistor. In the absence of C_{in} the signal source will be in parallel with resistor R_2 and the bias will be affected. Thus the function of C_{in} is to allow only the alternating current from the signal source to flow in to the input circuit. The emitter bypass capacitor C_E offers low reactance path to the signal. If it is not present, the voltage drop across R_E will reduce the effective voltage available across the base-emitter terminals (the input voltage) and thus reduces the gain. The coupling capacitor C_{out} transmits a.c signal but blocks the d.c voltage of the first stage from reaching the base of the second stage in the case of multistage amplifiers.

The emitter bypass capacitor **CE** offers low reactance path to the signal. If it is not present, the voltage drop across **RE** will reduce the effective voltage available across the base-emitter terminals (the input voltage) and thus reduces the gain.



Frequency response curve is a graph that indicates the relationship between voltage gain and function of frequency. The frequency response of a RC coupled amplifier is as shown in the following graph.



From the above graph, it is understood that the frequency rolls off or decreases for the frequencies below 50Hz and for the frequencies above 20 KHz. whereas the voltage gain for the range of frequencies between 50Hz and 20 KHz is constant.

At Low frequencies (i.e. below 50 Hz)

The capacitive reactance is inversely proportional to the frequency. At low frequencies, the reactance is quite high. The reactance of input capacitor C_{in} and the coupling capacitor C_c are so high that only small part of the input signal is allowed. The reactance of the emitter by pass capacitor C_E is also very high during low frequencies. Hence it cannot shunt the emitter resistance effectively. With all these factors, the voltage gain rolls off at low frequencies.

At High frequencies (i.e. above 20 KHz)

Again considering the same point, we know that the capacitive reactance is low at high frequencies. So, a capacitor behaves as a short circuit, at high frequencies. As a result of this, the loading effect of the next stage increases, which reduces the voltage gain. Along with this, as the capacitance of emitter diode decreases, it increases the base current of the transistor due to which the current gain (β) reduces. Hence the voltage gain rolls off at high frequencies.

At Mid-frequencies (i.e. 50 Hz to 20 KHz)

The voltage gain of the capacitors is maintained constant in this range of frequencies, as shown in figure. If the frequency increases, the reactance of the capacitor C_c decreases which tends to increase the gain. But this lower capacitance reactive increases the loading effect of the next stage by which there is a reduction in gain.

Due to these two factors, the gain is maintained constant.

Advantages of RC Coupled Amplifier

The following are the advantages of RC coupled amplifier.

- The frequency response of RC amplifier provides constant gain over a wide frequency range, hence most suitable for audio applications.
- The circuit is simple and has lower cost because it employs resistors and capacitors which are cheap.
- It becomes more compact with the upgrading technology.

Disadvantages of RC Coupled Amplifier

The following are the disadvantages of RC coupled amplifier.

- The voltage and power gain are low because of the effective load resistance.

- They become noisy with age.
- Due to poor impedance matching, power transfer will be low.

Applications of RC Coupled Amplifier

The following are the applications of RC coupled amplifier.

- They have excellent audio fidelity over a wide range of frequency.
- Widely used as Voltage amplifiers
- Due to poor impedance matching, RC coupling is rarely used in the final stages.

8 b) Narrate how capacitor filter eliminates ripples from the output of a rectifier. (4)

The output of a rectifier is pulsating and contains a steady DC component with undesirable ripples. If such pulsating DC is given to the electronic circuits, it produces disturbances and other interferences. Hence ripples have to be kept far from the load. This is achieved by use of a filter circuit in between the rectifier and load. The capacitor will be charged to the highest rectifier voltage value that is denoted with V_m , and then the voltage of the rectifier starts to reduce. As this happens, the capacitor starts discharging through the voltage across it and load. The voltage across the load will reduce little only because the next peak voltage occurs instantaneously to charge the capacitor. This procedure will repeat many times and the output waveform will be seen that very slight ripple is missing in the output. Furthermore, the output voltage is superior because it remains significantly close to the highest value of the output voltage of the rectifier.

9. a) What is the need of biasing? Draw the potential divider biasing circuit.

For normal operation, base-emitter junction should be forward biased and the collector-base junction reverse biased. The amount of bias required is significant for the establishment of the operating or the Q-point which is dictated by the mode of operation desired. In case the transistor is not biased properly, it would :

- work inefficiently
- produce distortion in the output signal
- with the change in transistor parameters or temperature rise, the operating point may shift and the amplifier output will be unstable.

A *potential divider bias*, also known as voltage divider bias, is a method used for the dc biasing of *bipolar junction transistors* (BJT) in a simple amplifier circuit. The circuit usually consists of biasing resistors in a voltage divider network whose values are determined through circuit analysis.

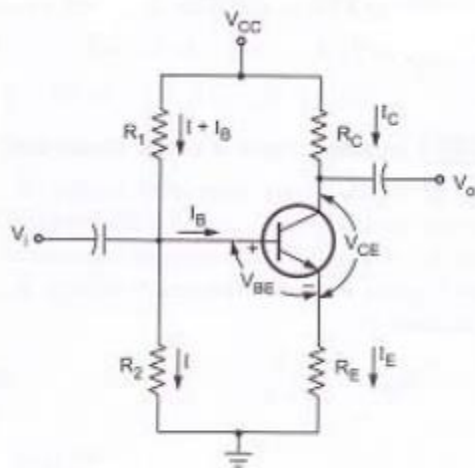


Fig. Voltage divider bias circuit

In the circuit above, R_1 and R_2 form a voltage divider network. This type of network has the effect of offsetting the input signal. Instead of swinging around 0 V, the signal then swings along a positive DC offset voltage. The resistors R_1 and R_2 determine this offset voltage.

Due to the biasing voltage at the base junction of the transistor, the transistor will always conduct slightly and consequently there will be a current I_C flowing even when there is no signal at the base. This current, known as *zero-signal collector current*, represents the proper offset required to push the signal into the positive half of the transistors operation.

9 (b) Explain the working of simple zener voltage Regulator.

The **Zener Diode** is used in its “reverse bias” or reverse breakdown mode, i.e. the diodes anode connects to the negative supply. From the I-V characteristics curve above, we can see that the zener diode has a region in its reverse bias characteristics of almost a constant negative voltage regardless of the value of the current flowing through the diode.

This voltage remains almost constant even with large changes in current providing the zener diodes current remains between the breakdown current $I_{Z(\min)}$ and its maximum current rating $I_{Z(\max)}$.

This ability of the zener diode to control itself can be used to great effect to regulate or stabilise a voltage source against supply or load variations. The fact that the voltage across the diode in the breakdown region is almost constant turns out to be an important characteristic of the zener diode as it can be used in the simplest types of voltage regulator applications.

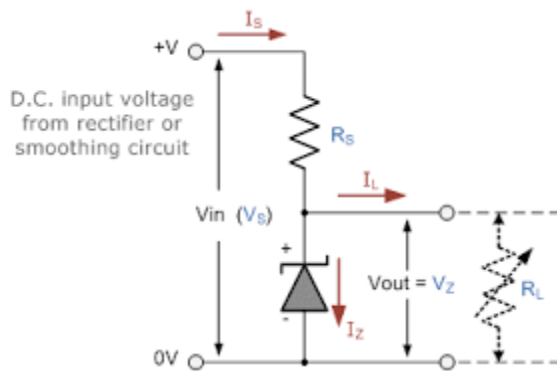
The function of a voltage regulator is to provide a constant output voltage to a load connected in parallel with it in spite of the ripples in the supply voltage or variations in the load current. A zener diode will continue to regulate its voltage until the diodes holding current falls below the minimum $I_{Z(\min)}$ value in the reverse breakdown region.

The Zener Diode Regulator

Zener Diodes can be used to produce a stabilised voltage output with low ripple under varying load current conditions. By passing a small current through the diode from a voltage source, via a suitable current limiting resistor (R_S), the zener diode will conduct sufficient current to maintain a voltage drop of V_{out} .

The DC output voltage from the half or full-wave rectifiers contains ripple superimposed onto the DC voltage and that as the load value changes so to does the average output voltage. By connecting a simple zener stabiliser circuit as shown below across the output of the rectifier, a more stable output voltage can be produced.

Resistor, R_S is connected in series with the zener diode to limit the current flow through the diode with the voltage source, V_S being connected across the combination. The stabilised output voltage V_{out} is taken from across the zener diode.



The zener diode is connected with its cathode terminal connected to the positive rail of the DC supply so it is reverse biased and will be operating in its breakdown condition. Resistor R_S is selected so to limit the maximum current flowing in the circuit.

With no load connected to the circuit, the load current will be zero, ($I_L = 0$), and all the circuit current passes through the zener diode which in turn dissipates its maximum power. Also a small value of the series resistor R_S will result in a greater diode current when the load resistance R_L is connected and large as this will increase the power dissipation requirement of the diode so care must be taken when selecting the appropriate value of series resistance so that the zener's maximum power rating is not exceeded under this no-load or high-impedance condition.

The load is connected in parallel with the zener diode, so the voltage across R_L is always the same as the zener voltage, ($V_R = V_Z$). There is a minimum zener current for which the stabilisation of the voltage is effective and the zener current must stay above this value operating under load within its breakdown region at all times. The upper limit of current is of course dependent upon the power rating of the device. The supply voltage V_S must be greater than V_Z .

One small problem with zener diode stabiliser circuits is that the diode can sometimes generate electrical noise on top of the DC supply as it tries to stabilise the voltage. Normally this is not a problem for most applications but the addition of a large value decoupling capacitor across the zener's output may be required to give additional smoothing.

Then to summarise a little. A zener diode is always operated in its reverse biased condition. As such a simple voltage regulator circuit can be designed using a zener diode to maintain a constant DC output voltage across the load in spite of variations in the input voltage or changes in the load current.

10.a) What are the merits of AM compared to FM. The carrier amplitude of a given AM wave is 5V and the message signal amplitude is 3V. Find the modulation index.

A.M	F.M
A.M is used in wide range of frequencies.	Range of F.M is restricted.
Narrow bandwidth is required.	Large bandwidth is required.
The equipment required for A.M transmission is less costly.	The equipment required is complex and also expensive.
Most of power in A.M is in the carrier which does not carry any information.	All the power transmitted is useful in the F.M system.

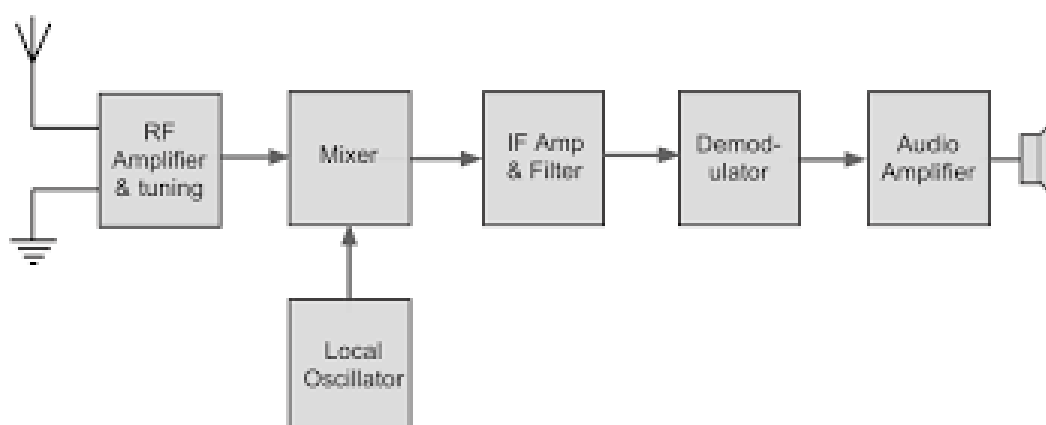
Signal to noise ratio is poor in A.M.	Signal to noise ratio is more in F.M.
Interferences are more.	Interferences are less.

Modulation index is a term used to describe the amount of amplitude change present in an AM waveform .It is the ratio of amplitude of the modulating signal to the amplitude of the carrier signal.

$$m = E_m/E_c$$

$$m = 3/5 = 0.6$$

10 b) Explain the block diagram of heterodyne receiver.



RF tuning & amplification: This RF stage within the overall block diagram for the receiver provides initial tuning to remove the image signal. It also provides some amplification. There are many different approaches used within the RF circuit design for this block dependent its application.

The RF circuit design presents some challenges. Low cost broadcast radios may have an amplifying mixer circuit that gives some RF amplification. HF radios may not want too much RF gain because some of the very strong signals received could overload later stages. The RF design may incorporate some amplification as well as RF attenuation to overcome this issue. Radios for VHF and above will tend to use more gain to have a noise figure that is sufficiently low to receive the signal. Noise is a particular problem for VHF / UHF radio communications systems.

If noise performance for the receiver is important, then this stage will be designed for optimum noise performance. This RF amplifier circuit block will also increase the signal level so that the noise introduced by later stages is at a lower level in comparison to the wanted signal.

All radios will need a sufficiently high level of image rejection, and this is provided by the RF tuning. High IF frequencies enable the RF tuning to be more effective as the difference between the wanted signal and image is increased.

- *Local oscillator:* Like other areas of the RF circuit design, the local oscillator circuit block within the superhet radio can take a variety of forms.

Most receivers use one or more of a variety of forms frequency synthesizers. The most common approach in the RF circuit design is to use a phase locked loop approach. Single and multi-loop synthesizers are used dependent upon the requirements, performance, cost and the like. Direct digital synthesizers are also being used increasingly.

Mixer: The mixer can be one of the key elements within the overall RF design of the receiver. Ensuring that the mixer performance matches that of the rest of the radio is particularly important.

Both the local oscillator and incoming signal enter this block within the superheterodyne receiver. The wanted signal is converted to the intermediate frequency.

For a high performance radio used for two way radio communications and the like, much better performance is required. To achieve this mixer circuits such as balanced mixers, double balanced mixers, and the like may be seen

IF amplifier & filter: This superheterodyne receiver block provides the majority of gain and selectivity. Often comparatively little gain will be provided in the previous blocks of the RF circuit design of the radio. The IF stages are where the main gain is provided. Being fixed in frequency, it is much easier to achieve high levels of gain and overall performance.

The IF stage might have included a number of different transistors, FETs or thermionic valves / vacuum tubes and other electronic components, but nowadays it is possible to obtain integrated circuits that contain a complete IF strip.

- *Demodulator:* The superheterodyne receiver block diagram only shows one demodulator, but in reality many radio RF designs may have one or more demodulators dependent upon the type of signals being received. Those radios used for professional radio communications applications and monitoring may need to be able to demodulate a variety of modulation schemes and waveforms and this may require a number of different demodulators that can be switched in as appropriate.
- *Automatic Gain Control, AGC:* An automatic gain control is incorporated into most superhet radio block diagrams. The function of this circuit block is to reduce the gain for strong signals so

that the audio level is maintained for amplitude sensitive forms of modulation, and also to prevent overloading.

Although the basic concept is the same through all radio RF circuit designs, there are some variations in the implementation. Some of the key variations are the time constant of the AGC system. For AM and the like a relatively slow time constant is acceptable. For SSB, a shorter time constant is needed so that the envelope of the SSB signal is followed.

There are also variations in the way the AGC voltage is derived, and where it is applied. Often it is applied to the IF circuit blocks first and then to the RF circuit block. In this way the best signal to noise ratio is preserved. Generally the AGC is relatively easy to implement, having relatively few electronic components.

Audio amplifier: Once demodulated, the recovered audio is applied to an audio amplifier block to be amplified to the required level for loudspeakers or headphones. Alternatively the recovered modulation may be used for other applications whereupon it is processed in the required way by a specific circuit block.

In many ways, this circuit block within the superheterodyne radio is the most straightforward. For many applications, the audio amplifier will involve some straightforward electronic circuit design, especially if the audio is applied to simple headphones or a loudspeaker. For two way radio communication applications, the audio bandwidth may need to be limited to the "telecommunications" bandwidth of about 300 Hz to 3.3 kHz. Audio filters could be employed as well.

Block diagram explanation

Signals enter the receiver from the antenna and are applied to the RF amplifier where they are tuned to remove the image signal and also reduce the general level of unwanted signals on other frequencies that are not required.

The signals are then applied to the mixer along with the local oscillator where the wanted signal is converted down to the intermediate frequency. Here significant levels of amplification are applied and the signals are filtered. This filtering selects signals on one channel against those on the next. It is much larger than that employed in the front end. The advantage of the IF filter as opposed to RF filtering is that the filter can be designed for a fixed frequency. This allows for much better tuning. Variable filters are never able to provide the same level of selectivity that can be provided by fixed frequency ones.

Once filtered the next block in the superheterodyne receiver is the demodulator. This could be for amplitude modulation, single sideband, frequency modulation, or indeed any form of modulation. It is also possible to switch different demodulators in according to the mode being received.

The final element in the superheterodyne receiver block diagram is shown as an audio amplifier, although this could be any form of circuit block that is used to process or amplify the demodulated signal.

11. a) Describe the principle of antenna.

An antenna is basically a length of a conductor that acts as a conversion device. It serves as a link between the transmitter and the receiver. An antenna is a transducer, which converts electrical power into electromagnetic wave and vice versa. An antenna is also termed as Aerial. It

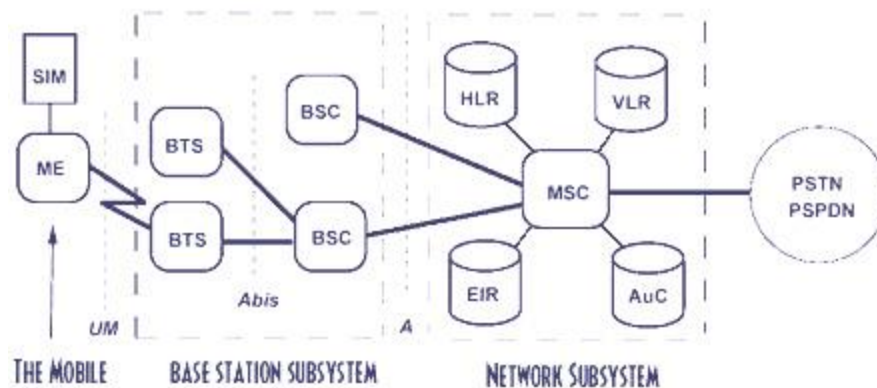
can be used to both transmit and receive electromagnetic waves and obeys reciprocity principle.

b) With necessary block diagram explain the working of GSM.

GSM (Global System for Mobile communications: originally from Groupe Spécial Mobile) is the most popular standard for mobile phones in the world. GSM differs from its predecessors in that both signaling and speech channels are digital, and thus is considered a second generation (2G) mobile phone system. This has also meant that data communication was easy to build into the system.

The different features of GSM

- **Short message service (SMS)**, which is a bidirectional service for sending alphanumeric message up to 160 bytes in length.
- **Subscriber identity module (SIM)**, which is a memory device that stores information such as subscriber's identification number, the networks and countries where the subscriber is entitled to service, privacy keys, and other user specific information.
- **On-the-air privacy** which is provided by the system. The privacy is made possible by encrypting the digital bit stream sent by a GSM transmitter.



SIM – Subscriber Identification Module.

MS – Mobile Station.

ME – Mobile Equipment.

BTS – Bare Transceiver System.

MSC – Mobile Services switching Center.

NSS – Network & Switching Subsystem.

PSTN – Public Switched Telephone Network.

User Equipment (UE)—These are the users .Number of users are controlled by one BTS

1. The mobile stations (MS) communicate with the base station subsystem over the radio the radio interface.
2. The BSS called as radio the subsystem, provides and manages the radio transmission path between the mobile stations and the Mobile Switching Centre(MSC).It also manages radio interface between the mobile stations and other subsystems of GSM.
3. Each BSS comprises many Base Station Controllers(BSC) that connect the mobile station to the network and switching subsystem (NSS) through the mobile switching center

4. The NSS controls the switching functions of the GSM system. It allows the mobile switching center to communicate with networks like PSTN, ISDN, CSPDN, PSPDN and other data networks.
5. The operation support system (OSS) allows the operation and maintenance of the GSM system. It allows the system engineers to diagnose, troubleshoot and observe the parameters of the GSM systems. The OSS subsystem interacts with the other subsystems and is provided for the GSM operating company staff that provides service facilities for the network.

Base station(BSS)-- The following stations subsystem comprises of two parts:

1. Base Transceiver Station (BTS).
2. Base Station Controller(BSC).

The BSS consists many BSC that connect to a single MSC. Each BSC controls up to several hundred BTS.

Base Transceiver Station(BTS)-BTS

It has radio transceiver that define a cell and are capable of handling radio link protocols with MS.

Functions of BTS are

1. Handling radio link protocols
2. Providing FD communication to MS.
3. Interliving and de- interliving.

Base station controller(BSC) It manages radio resources for one or more BTS. It controls several hundred BTS all are connected to single MSC.

Functions of BSC are

- To control BTS.
- Radio resource management
- Handoff management and control
- Radio channel setup and frequency hopping

Network subsystem(NSS)

1. It handles the switching of GSM calls between external networks and indoor BSC
2. It includes three different data bases for mobility management as
 - A .HLR (Home Location Register)
 - B .VLR (Visitor Location Register)
 - C. AUC (Authentication center)

Mobile switching center (MSC)--

It connects fix networks like ISDN ,PSTN etc.

Following are the functions of MSC

1. Call setup, supervision and release
2. COLLECTION OF BILLING INFORMATION
3. Call handling / routing
4. Management of signalling protocol
5. Record of VLR and HLR

HLR (Home Location Register) - Call roaming and call routing capabilities of GSM are handled. It stores all the administrative information of subscriber registered in the networks. It maintains unique international mobile subscriber identity.(IMSI).

VLR (Visitor Location Register) - It is a temporary data base. It stores the IMSI number and customer information for each roaming customer visiting specific MSC.

Authentication center - It is protected database .It maintains authentication keys and algorithms.It contains a register called as Equipment Identity Register.

Operation subsystem(OSS) - IT manages all mobile equipment in the system
1)management for charging and billing procedure 2)To maintain all hardware and network operations

Interfaces used for GSM network :

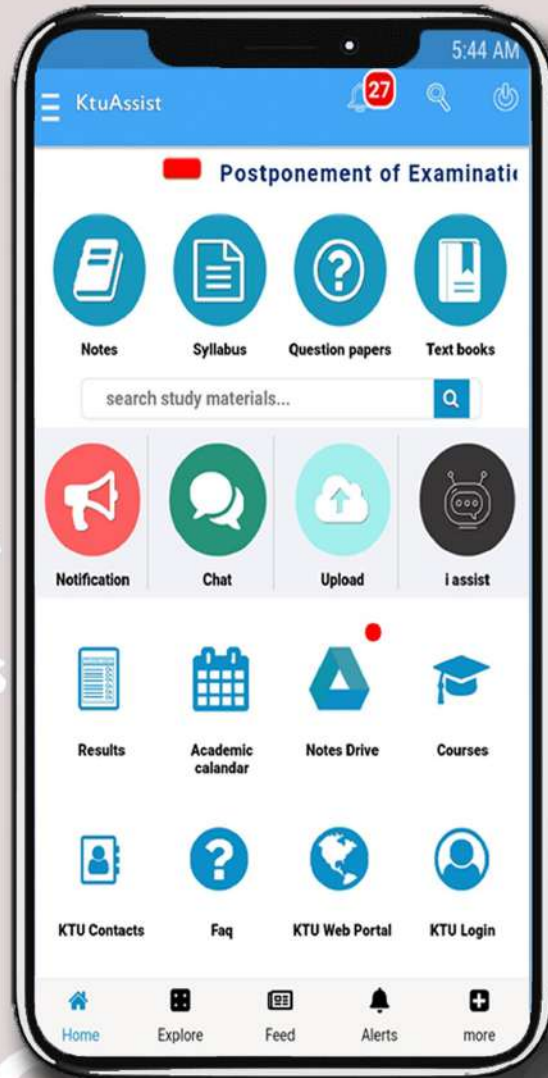
- 1)UM Interface –Used to communicate between BTS with MS
- 2)Abis Interface— Used to communicate BSC TO BTS
- 3)A Interface-- Used to communicate BSC and MSC
- 4) Singling protocol (SS 7)- Used to communicate MSC with other network .

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